Practical Journal

In

Microservice Architecture

Submitted By

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PART – I

DEPARTMENT OF INFORMATION TECHNOLOGY

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KISHINCHAND CHELLARAM COLLEGE

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DEPARTMENT OF INFORMATION TECHNOLOGY

M.Sc.IT PART- I

CERTIFICATE

This is to certify that the Practical conducted by Mr **Prem Patil** for M.Sc. (IT) Part- I Semester- II, Seat No: **25FMSCIT-015** at Kishinchand Chellaram College in partial fulfilment for the MASTERS OF SCIENCE (INFORMATION TECHNOLOGY). Degree Examination for semester II has been periodically examined and signed, and the course of term work has been satisfactorily carried out for the year 2025 - 2026. This Practical journal has not been submitted for any other examination and does not form part of any other course undergone by the candidate.

Signature	**Signature**	**Signature**
Lecturer-In-Charge	**External Examiner**	**Course Coordination**
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College Stamp

SUBJECT CODE –

Microservice Architecture

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3	Creating a glossary Web Api (swagger).		
4	Working with docker, creating/building docker image and then storing it in a container.		
5	Learn how to build a docker image using a docker file run a container and push the image to docker hub using docker desktop.		
6	Working with docker volumes & network.		
7	Working with docker update commands.		
8	Working with Circle CI for continuous integration.		

Practical No. 1

Aim: Implement a microservices based system where independent services (user services & ride services) communicate with each other using REST APIs demonstrating loose coupling independent deployment and services-to-services interaction.

Step 1: Check python version in cmd.

```
PS C:\Users\hutox> py -0
 -V:3.14 *        Python 3.14 (64-bit)
 -V:3.10          Python 3.10 (64-bit)
PS C:\Users\hutox>
```

Use python 3.10 or any version higher than 3 explicitly.

Step 2: Install required libraries in cmd.

```
PS C:\Users\hutox> pip install flask
Requirement already satisfied: flask in .\appdata\local\programs\python\python310\lib\site-packages (3
Requirement already satisfied: blinker>=1.9.0 in .\appdata\local\programs\python\python310\lib\site-pa
from flask) (1.9.0)
Requirement already satisfied: click>=8.1.3 in .\appdata\local\programs\python\python310\lib\site-pack
om flask) (8.1.8)
Requirement already satisfied: itsdangerous>=2.2.0 in .\appdata\local\programs\python\python310\lib\si
ges (from flask) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in .\appdata\local\programs\python\python310\lib\site-pac
rom flask) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in .\appdata\local\programs\python\python310\lib\site
s (from flask) (3.0.3)
Requirement already satisfied: werkzeug>=3.1.0 in .\appdata\local\programs\python\python310\lib\site-p
(from flask) (3.1.5)
Requirement already satisfied: colorama in .\appdata\local\programs\python\python310\lib\site-packages
lick>=8.1.3->flask) (0.4.6)
PS C:\Users\hutox> pip install requests
Requirement already satisfied: requests in .\appdata\local\programs\python\python310\lib\site-packages
)
Requirement already satisfied: charset_normalizer<4,>=2 in .\appdata\local\programs\python\python310\l
packages (from requests) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in .\appdata\local\programs\python\python310\lib\site-pack
om requests) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in .\appdata\local\programs\python\python310\lib\sit
es (from requests) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in .\appdata\local\programs\python\python310\lib\sit
es (from requests) (2026.1.4)
PS C:\Users\hutox>
```

Step 3: Create a folder. Inside it create user_service.py file.

```
from flask import Flask, jsonify

app = Flask(__name__)

@app.route('/user', methods=['GET'])
def get_user():
    return jsonify({
        "user_id": 101,
        "name": "Prem Patil",
        "role": "Rider"
    })

if __name__ == '__main__':
    app.run(port=5001)
```

Run this file:

```
========= RESTART: C:/Users/hutox/Desktop/MA Prac
 * Serving Flask app 'user_service'
 * Debug mode: off
[31m[1mWARNING: This is a development server. Do
ployment. Use a production WSGI server instead.[(
 * Running on http://127.0.0.1:5001
[33mPress CTRL+C to quit[0m
127.0.0.1 - - [08/Apr/2026 15:57:59] "[33mGET / H
127.0.0.1 - - [08/Apr/2026 15:58:32] "GET /user H
```

Output:

```
{
    "name": "Prem Patil",
    "role": "Rider",
    "user_id": 101
}
```

Step 4: Create another ride service & save it as ride_service inside same folder.

```
from flask import Flask, jsonify
import requests

app = Flask(__name__)

@app.route('/ride',methods=['GET'])
def create_ride():
    try:
        response = requests.get("http://localhost:5001/user")
        user = response.json()


        return jsonify({
            "ride_id": 5001,
            "user": user["name"],
            "status": "Ride Booked"
    })

    except Exception as e:
        return jsonify({"error":str(e)}),500

if __name__ == '__main__':
    app.run(port=5002)
```

```
 * Serving Flask app 'ride_service'
 * Debug mode: off
[31m[1mWARNING: This is a development server. D
ployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5002
[33mPress CTRL+C to quit[0m
127.0.0.1 - - [08/Apr/2026 16:03:02] "GET /ride
127.0.0.1 - - [08/Apr/2026 16:03:02] "[33mGET /
```

```
{
    "ride_id": 5001,
    "status": "Ride Booked",
    "user": "Prem Patil"
}
```

Practical No. 2

Aim: ASP.NET CORE MVC.

Step 1: Install .NET core sdk: https://dotnet.microsoft.com/learn/dotnet/hello-world-tutorial/install

```
C:\Users\hutox\Desktop\MA Practical>dotnet

Usage: dotnet [path-to-application]
Usage: dotnet [commands]

path-to-application:
  The path to an application .dll file to execute.

commands:
  -h|--help                         Display help.
  --info                            Display .NET information.
  --list-runtimes [--arch <arch>]   Display the installed runtimes matching
4, x86.
  --list-sdks [--arch <arch>]       Display the installed SDKs matching the
86.

C:\Users\hutox\Desktop\MA Practical>dotnet --version
10.0.201

C:\Users\hutox\Desktop\MA Practical>
```

Step 2: Create folder MyMVC and Run dotnet new mvc.

```
C:\Users\hutox\Desktop\MA Practical\MyMVC>dotnet new mvc

Welcome to .NET 10.0!
---------------------
SDK Version: 10.0.201

Telemetry
---------
The .NET tools collect usage data in order to help us improve your experience. It is c
an opt-out of telemetry by setting the DOTNET_CLI_TELEMETRY_OPTOUT environment variabl

Read more about .NET CLI Tools telemetry: https://aka.ms/dotnet-cli-telemetry

----------------
Installed an ASP.NET Core HTTPS development certificate.
To trust the certificate, run 'dotnet dev-certs https --trust'
Learn about HTTPS: https://aka.ms/dotnet-https

----------------
Write your first app: https://aka.ms/dotnet-hello-world
Find out what's new: https://aka.ms/dotnet-whats-new
Explore documentation: https://aka.ms/dotnet-docs
Report issues and find source on GitHub: https://github.com/dotnet/core
Use 'dotnet --help' to see available commands or visit: https://aka.ms/dotnet-cli
--------------------------------------------------------------------------------
The template "ASP.NET Core Web App (Model-View-Controller)" was created successfully.
This template contains technologies from parties other than Microsoft, see https://aka

Processing post-creation actions...
Restoring C:\Users\hutox\Desktop\MA Practical\MyMVC\MyMVC.csproj:
Restore succeeded.
```

Step 3: Go to controller folder and go to HomeController.cs file modify and match the following.

```
using System;
using System.Collections.Generic;
using System.Diagnostics;
using System.Linq;
using System.Threading.Tasks;
using Microsoft.AspNetCore.Mvc;
using Microsoft.Extensions.Logging;
using MyMVC.Models;

namespace MyMVC.Controllers
{
    public class HomeController : Controller
    {
        public string Index()
        {
            return "Prem Patil";
        }
    }
}
```

Step 4: Run the project.

```
C:\Users\hutox\Desktop\MA Practical\MyMVC>dotnet run
Using launch settings from C:\Users\hutox\Desktop\MA Practical\MyMVC\P
Building...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5037
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\hutox\Desktop\MA Practical\MyMVC
warn: Microsoft.AspNetCore.HttpsPolicy.HttpsRedirectionMiddleware[3]
      Failed to determine the https port for redirect.
```

Step 5: Stop running project by pressing CTL+C.

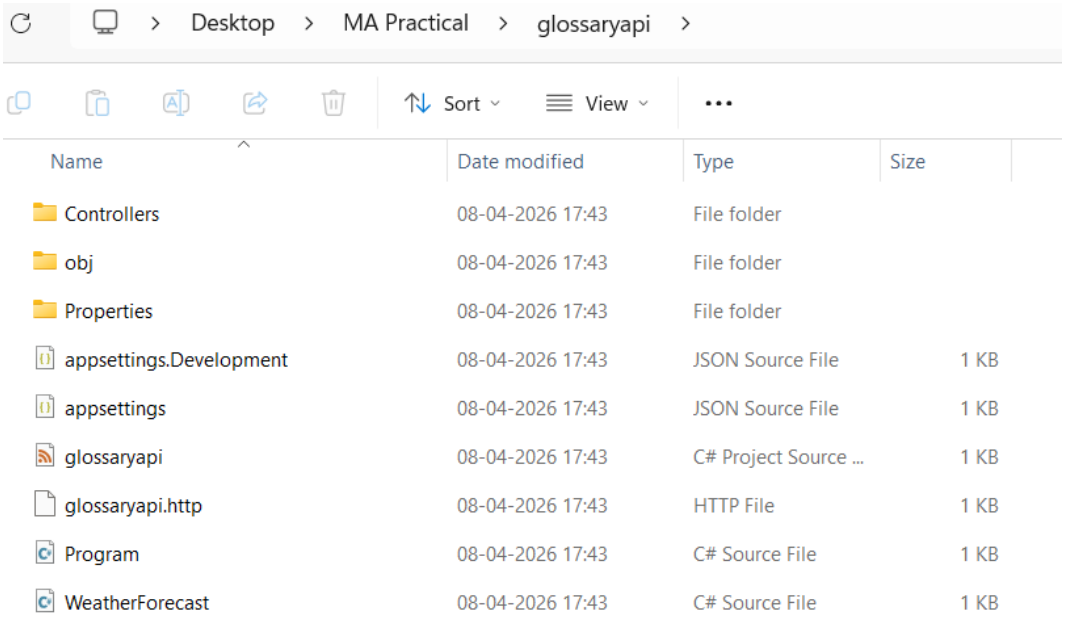
```
C:\Users\hutox\Desktop\MA Practical\MyMVC>dotnet run
Using launch settings from C:\Users\hutox\Desktop\MA Practical\
Building...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5037
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\hutox\Desktop\MA Practical\My
warn: Microsoft.AspNetCore.HttpsPolicy.HttpsRedirectionMiddlewa
      Failed to determine the https port for redirect.
info: Microsoft.Hosting.Lifetime[0]
      Application is shutting down...

C:\Users\hutox\Desktop\MA Practical\MyMVC>
```

Practical No. 3

Aim: Creating a WebAPI in .NET.

Step 1: Create new .NET WebAPI.

```
C:\Users\hutox\Desktop\MA Practical>dotnet new webapi --use-controllers -n glossaryapi
The template "ASP.NET Core Web API" was created successfully.

Processing post-creation actions...
Restoring C:\Users\hutox\Desktop\MA Practical\glossaryapi\glossaryapi.csproj:
Restore succeeded.
```



Step2: Add package Swashbuckle.AspNetCore

```
C:\Users\hutox\Desktop\MA Practical\glossaryapi>dotnet add package Swashbuckle.AspNetCore --version 7.0.0
info : X.509 certificate chain validation will use the default trust store selected by .NET for code signing.
info : X.509 certificate chain validation will use the default trust store selected by .NET for timestamping.
info : Adding PackageReference for package 'Swashbuckle.AspNetCore' into project 'C:\Users\hutox\Desktop\MA Pr
ical\glossaryapi\glossaryapi.csproj'.
info : Restoring packages for C:\Users\hutox\Desktop\MA Practical\glossaryapi\glossaryapi.csproj...
info :   GET https://api.nuget.org/v3-flatcontainer/swashbuckle.aspnetcore/index.json
info :   OK https://api.nuget.org/v3-flatcontainer/swashbuckle.aspnetcore/index.json 601ms
info :   GET https://api.nuget.org/v3-flatcontainer/swashbuckle.aspnetcore/7.0.0/swashbuckle.aspnetcore.7.0.0.
kg
```

Step 3: Add the following code in Program.cs.

glossaryapi > Program.cs

```
var builder = WebApplication.CreateBuilder(args);

// Add services to the container.

builder.Services.AddControllers();
// Learn more about configuring OpenAPI at https://aka.ms/aspnet/openapi
builder.Services.AddOpenApi();

builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();

var app = builder.Build();

// Configure the HTTP request pipeline.
if (app.Environment.IsDevelopment())
{
    app.MapOpenApi();
}
app.UseSwagger();
app.UseSwaggerUI();
app.UseHttpsRedirection();
app.UseAuthorization();
app.MapControllers();
app.Run();

```

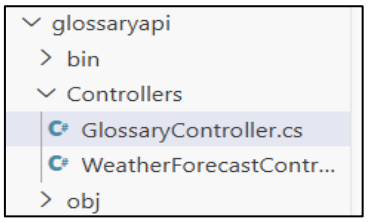
Step 4: Create a file GlossaryItem.cs and add the following code in it ().

MSC_MICROSERVICES
- glossaryapi
  - bin
  - Controllers
    - WeatherForecastContr...
  - obj
  - Properties
  - appsettings.Developmen...
  - appsettings.json
  - glossaryapi.csproj
  - glossaryapi.http
  - GlossaryItem.cs
  - Program.cs
  - WeatherForecast.cs

glossaryapi > GlossaryItem.cs

```
namespace glossaryapi {
    public class GlossaryItem {
        public int Id { get; set; }
        public string Term { get; set; }
        public string Definition { get; set; }
    }
}

```

Step 5: In controllers folder create a file GlossaryController.cs and add following code in it.



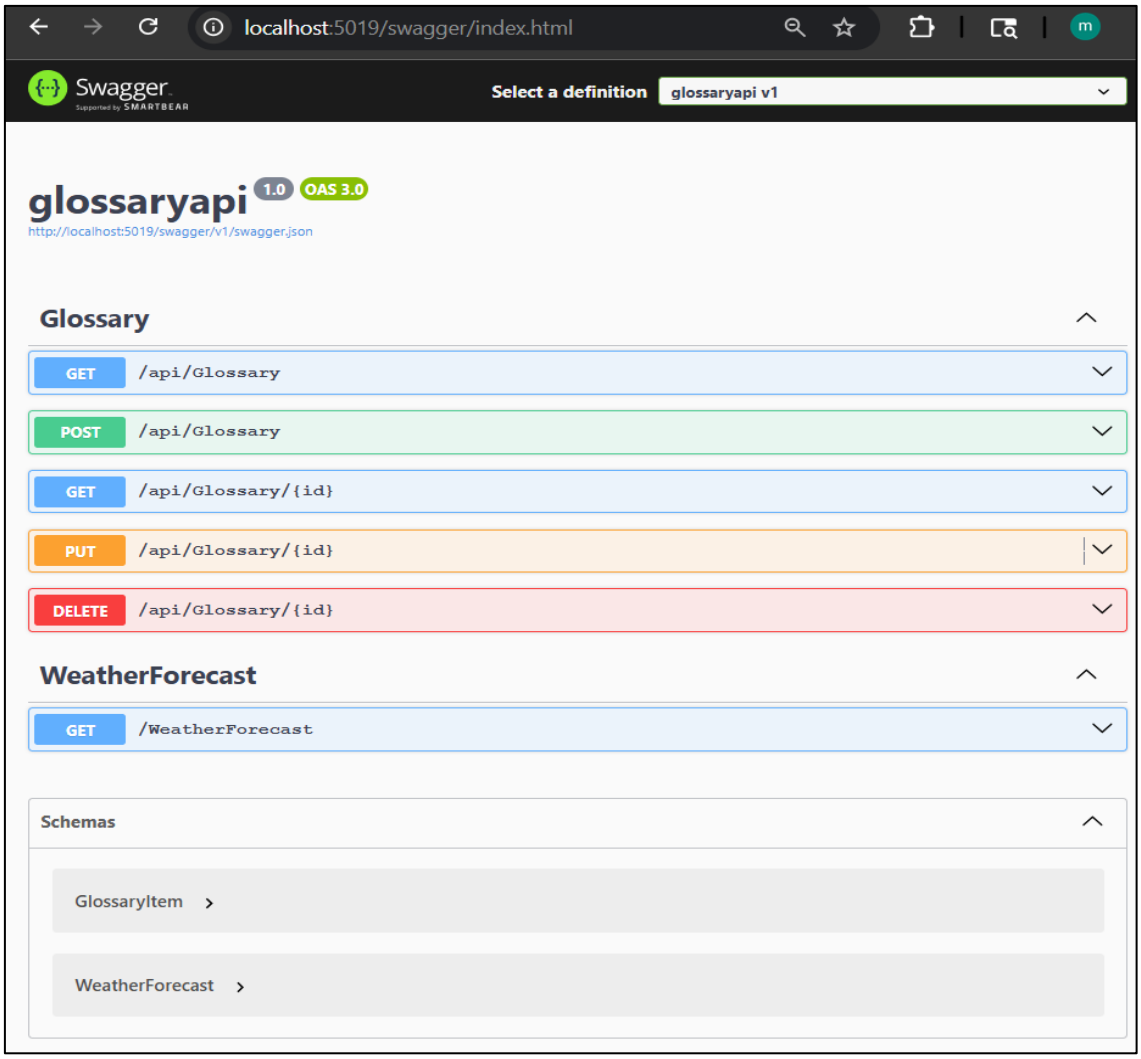
glossaryapi > Controllers > GlossaryController.cs

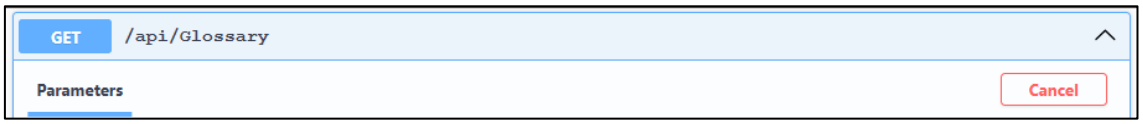
```
using Microsoft.AspNetCore.Mvc;
using System.Collections.Generic;
using System.Linq;
namespace glossaryapi.Controllers {
    [ApiController]
    [Route("api/[controller]")]
    public class GlossaryController : ControllerBase {
        private static List<GlossaryItem> glossary = new List<GlossaryItem> {
                new GlossaryItem { Id = 1, Term="API",
                    Definition="Application Programming Interface"},
                new GlossaryItem { Id = 2, Term="HTTP",
                    Definition="HyperText Transfer Protocol"}
        };
        [HttpGet]
        public ActionResult<List<GlossaryItem>> Get() {
            return Ok(glossary);
        }
        [HttpGet("{id}")]
        public IActionResult GetById(int id) {
            var item = glossary.FirstOrDefault(x => x.Id == id);
            return item == null ? NotFound() : Ok(item);
        }
        [HttpPost]
        public IActionResult Create(GlossaryItem item) {
            item.Id = glossary.Max(x => x.Id) + 1;
            glossary.Add(item);
            return Ok(item);
        }
        [HttpPut("{id}")]
        public IActionResult Update(int id, GlossaryItem updated) {
            var item = glossary.FirstOrDefault(x => x.Id == id);
            if (item == null) return NotFound();
            item.Term = updated.Term;
            item.Definition = updated.Definition;
            return Ok(item);
        }
        [HttpDelete("{id}")]
        public IActionResult Delete(int id) {
            var item = glossary.FirstOrDefault(x => x.Id == id);
            if (item == null) return NotFound();
            glossary.Remove(item);
            return Ok();
        }
    }
}
```

Step 6: Run the project.

```
E:\MITALI\PRACTICALS\msc__microservices\glossaryapi>dotnet run
Using launch settings from E:\MITALI\PRACTICALS\msc__microservices\glossaryapi\Prop
erties\launchSettings.json...
Building...
E:\MITALI\PRACTICALS\msc__microservices\glossaryapi\GlossaryItem.cs(4,23): warning
CS8618: Non-nullable property 'Term' must contain a non-null value when exiting con
structor. Consider adding the 'required' modifier or declaring the property as null
able.
E:\MITALI\PRACTICALS\msc__microservices\glossaryapi\GlossaryItem.cs(5,23): warning
CS8618: Non-nullable property 'Definition' must contain a non-null value when exiti
ng constructor. Consider adding the 'required' modifier or declaring the property a
s nullable.
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5019
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: E:\MITALI\PRACTICALS\msc__microservices\glossaryapi
```

Output:





No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:5019/api/Glossary' \
  -H 'accept: text/plain'
```

Request URL

```
http://localhost:5019/api/Glossary
```

Server response

Code	Details
200	Response body

```
[
  {
    "id": 1,
    "term": "API",
    "definition": "Application Programming Interface"
  },
  {
    "id": 2,
    "term": "HTTP",
    "definition": "HyperText Transfer Protocol"
  }
]
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Fri,03 Apr 2026 05:11:30 GMT
server: Kestrel
transfer-encoding: chunked
```

GET /api/Glossary/{id}

Parameters Cancel

Name	Description
id * required integer ($int32) (path)	1

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:5019/api/Glossary/1' \
  -H 'accept: */*'
```

Request URL

```
http://localhost:5019/api/Glossary/1
```

Server response

Code	Details
200	Response body

```
{
  "id": 1,
  "term": "API",
  "definition": "Application Programming Interface"
}
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Fri,03 Apr 2026 05:13:48 GMT
server: Kestrel
transfer-encoding: chunked
```

POST /api/Glossary

Parameters

Cancel Reset

No parameters

Request body application/json

```
{
  "id": 3,
  "term": "IP",
  "definition": "Internet Protocal"
}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:5019/api/Glossary' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "id": 3,
  "term": "IP",
  "definition": "Internet Protocal"
}'
```

Request URL

```
http://localhost:5019/api/Glossary
```

Server response

Server response

Code	Details
200	Response body

```
{
  "id": 3,
  "term": "IP",
  "definition": "Internet Protocal"
}
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Fri,03 Apr 2026 05:15:28 GMT
server: Kestrel
transfer-encoding: chunked
```

Responses

Code	Description	Links
200	OK	*No links*

Practical No. 4, 5

Aim: Working with docker creating/building docker image and then storing it in a container.

Step 1:

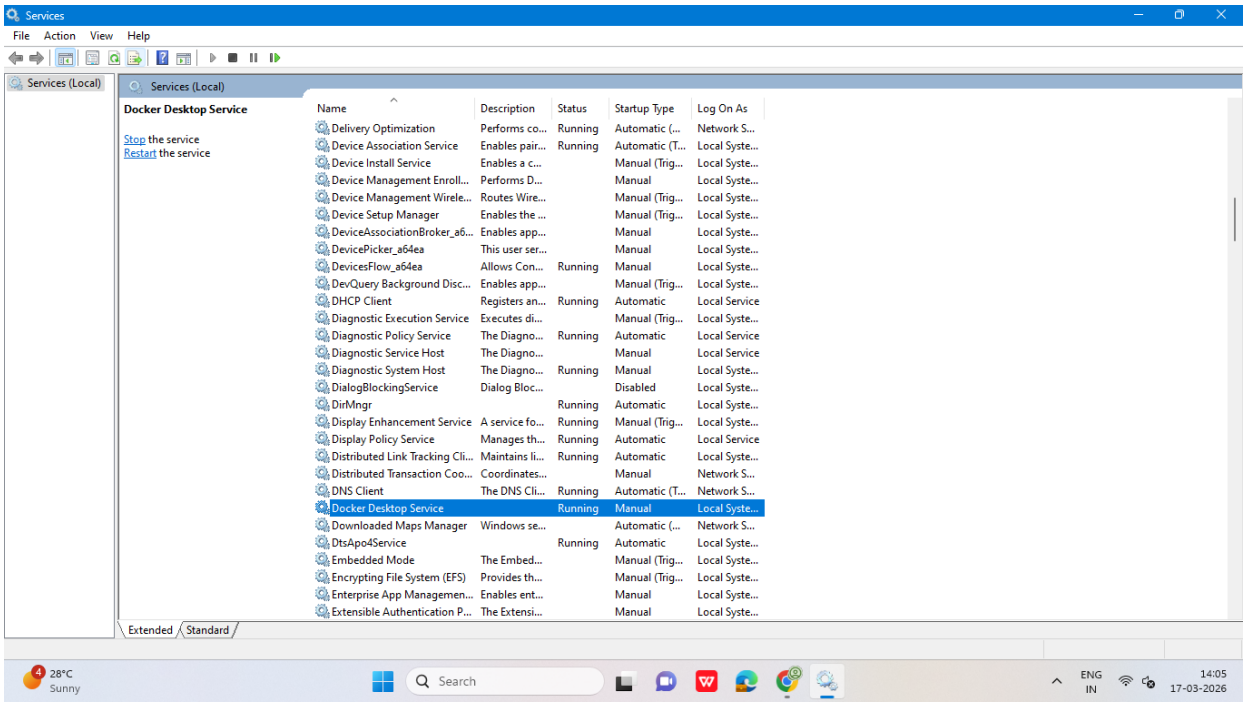
Install docker desktop install it on your system after installing restart the system.

https://www.docker.com/products/docker-desktop

Download docker hub https://hub.docker.com/

```
C:\Users\admin>docker --version
Docker version 29.2.1, build a5c7197

C:\Users\admin>docker info
Client:
 Version:    29.2.1
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: create or run AI agents (Docker Inc.)
    Version:  v1.29.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.19.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.32.1-desktop.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v5.1.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.3.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-desktop.exe
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.0-alpha
    Path:     C:\Program Files\Docker\cli-plugins\docker-dhi.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
```

Step 2: Go to system services and start the docker service.



Step 3: Download the busybox image from Docker Hub to your local machine

```
C:\Users\admin>docker pull busybox
Using default tag: latest
latest: Pulling from library/busybox
61dfb50712f5: Pull complete
96cfb76e59bd: Download complete
Digest: sha256:b3255e7dfbcd10cb367af0d409747d511aeb66dfac98cf30e97e87e4207dd76f
Status: Downloaded newer image for busybox:latest
docker.io/library/busybox:latest

C:\Users\admin>docker images

IMAGE            ID             DISK USAGE   CONTENT SIZE   EXTRA
busybox:latest   b3255e7dfbcd       6.77MB         2.22MB

C:\Users\admin>notepad Dockerfile
```

Step 4: Create Dockerfile.

```
C:\Users\admin>notepad Dockerfile
```

```
C:\Users\admin>ren Dockerfile.txt Dockerfile

C:\Users\admin>dir
 Volume in drive C has no label.
 Volume Serial Number is FA53-AFC4

 Directory of C:\Users\admin

17-03-2026  14:27    <DIR>          .
11-08-2025  11:40    <DIR>          ..
09-11-2025  13:46               676 #Simple Rule Based.py
12-09-2023  12:21    <DIR>          .arduinoIDE
17-03-2026  14:19    <DIR>          .aws
17-03-2026  14:19    <DIR>          .azure
12-03-2023  10:34           333,116 .babel.json
06-03-2026  11:03             2,196 .bash_history
```

Open the Dockerfile and add the following.

```
Dockerfile
File  Edit  View  H1  B

FROM busybox
```

Step 5: Build a Docker image

```
C:\Users\admin>docker build -t myfirstimage .
[+] Building 9.3s (5/5) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                1.7s
 => => transferring dockerfile: 101B                                                                0.4s
 => [internal] load metadata for docker.io/library/busybox:latest                                   0.2s
 => [internal] load .dockerignore                                                                   0.1s
 => => transferring context: 2B                                                                     0.0s
 => [1/1] FROM docker.io/library/busybox:latest@sha256:b3255e7dfbcd10cb367af0d409747d511aeb66dfac98cf30e97e87e4207dd76f  0.5s
 => => resolve docker.io/library/busybox:latest@sha256:b3255e7dfbcd10cb367af0d409747d511aeb66dfac98cf30e97e87e4207dd76f  0.1s
 => exporting to image                                                                              3.2s
 => => exporting layers                                                                             0.5s
 => => exporting manifest sha256:e190ea847e01e46187e03d380ab546364204cc8812bdbaa1bca35223f327cd44   0.0s
 => => exporting config sha256:6e6c8471cfa1dde45b9c0936305ee4e3fe44f07dad56624090e55f69f2df2e75     0.5s
 => => exporting attestation manifest sha256:67e2f6a57f14d405d8ca6d9ffb2a1d3b102161fd8b0c3d3e701b83e45fccfbb9  0.0s
 => => exporting manifest list sha256:43a9d254a58f7768d1cded73d75802ea811b06550df8553b49af4165531bc85f  0.0s
 => => naming to docker.io/library/myfirstimage:latest                                              0.0s
 => => unpacking to docker.io/library/myfirstimage:latest                                           0.4s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/rl5o5v1pdcv7l0h1qdfhevlb9
```

Containers Give feedback

Container CPU usage
No containers are running.

Container memory usage Show charts
No containers are running.

Search Only show running containers

Name	Container ID	Image	Port(s)	CPU (%)	Last started	Actions
brave_ptolemy	bdac4cb22843	myfirstimage		N/A	21 minutes ago	

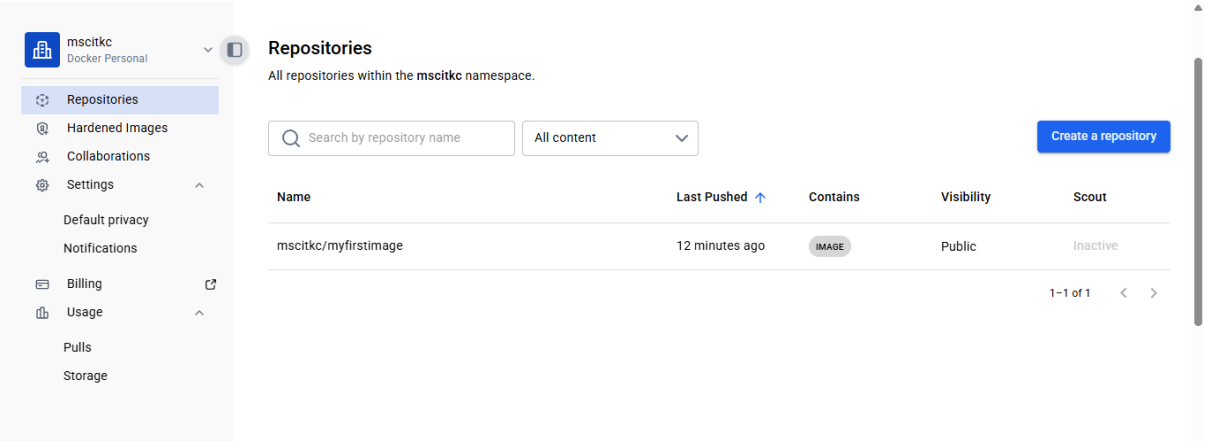
```
C:\Users\admin>docker images

IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
busybox:latest         b3255e7dfbcd       6.77MB         2.22MB
myfirstimage:latest    43a9d254a58f       6.77MB         2.21MB

C:\Users\admin>docker run myfirstimage
Hello! Docker practical is working
```

Step 6: Login to docker.

```
C:\Users\admin>docker login -u mscitkc

i Info → A Personal Access Token (PAT) can be used instead.
         To create a PAT, visit https://app.docker.com/settings


Password:

Login Succeeded
```

Step7: Then tag the image & push the image to the docker hub you will see the image is been added to docker hub & see them in docker desktop in repositories.

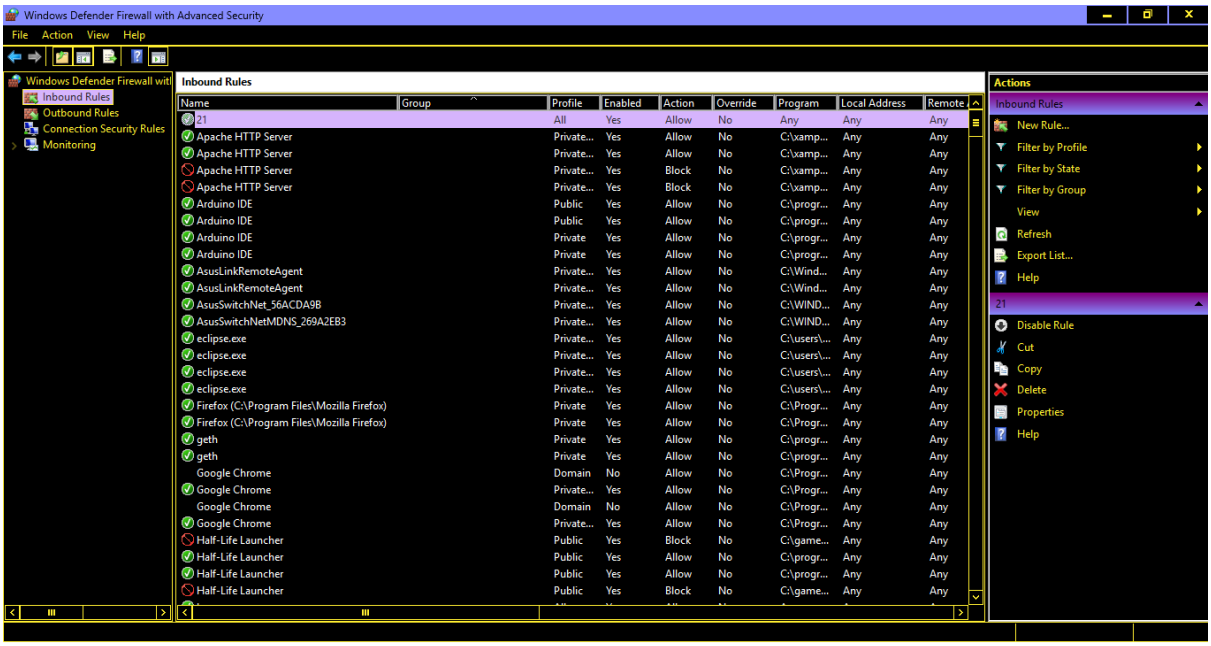
```
C:\Users\admin>docker tag myfirstimage mscitkc/myfirstimage:latest

C:\Users\admin>docker push mscitkc/myfirstimage:latest
The push refers to repository [docker.io/mscitkc/myfirstimage]
61dfb50712f5: Mounted from library/busybox
f3e819046b51: Pushed
latest: digest: sha256:43a9d254a58f7768d1cded73d75802ea811b06550df8553b49af4165531bc85f size: 855
```

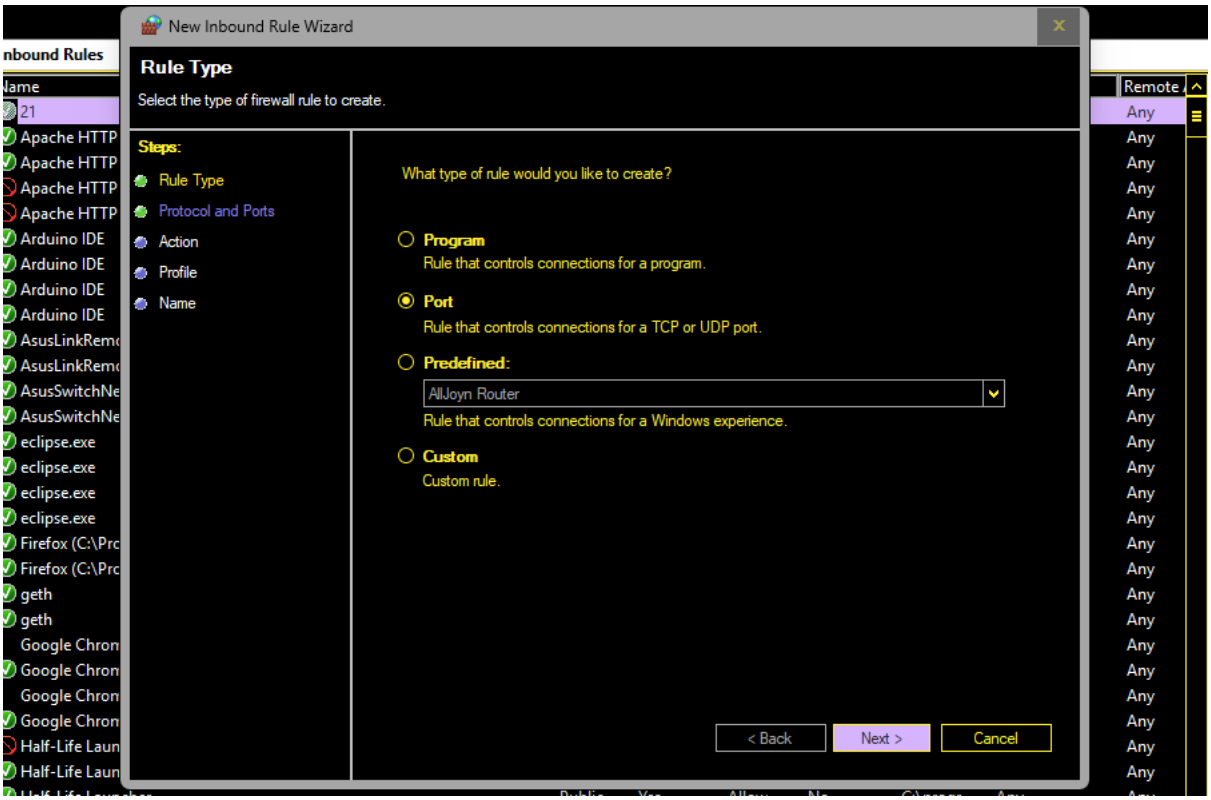


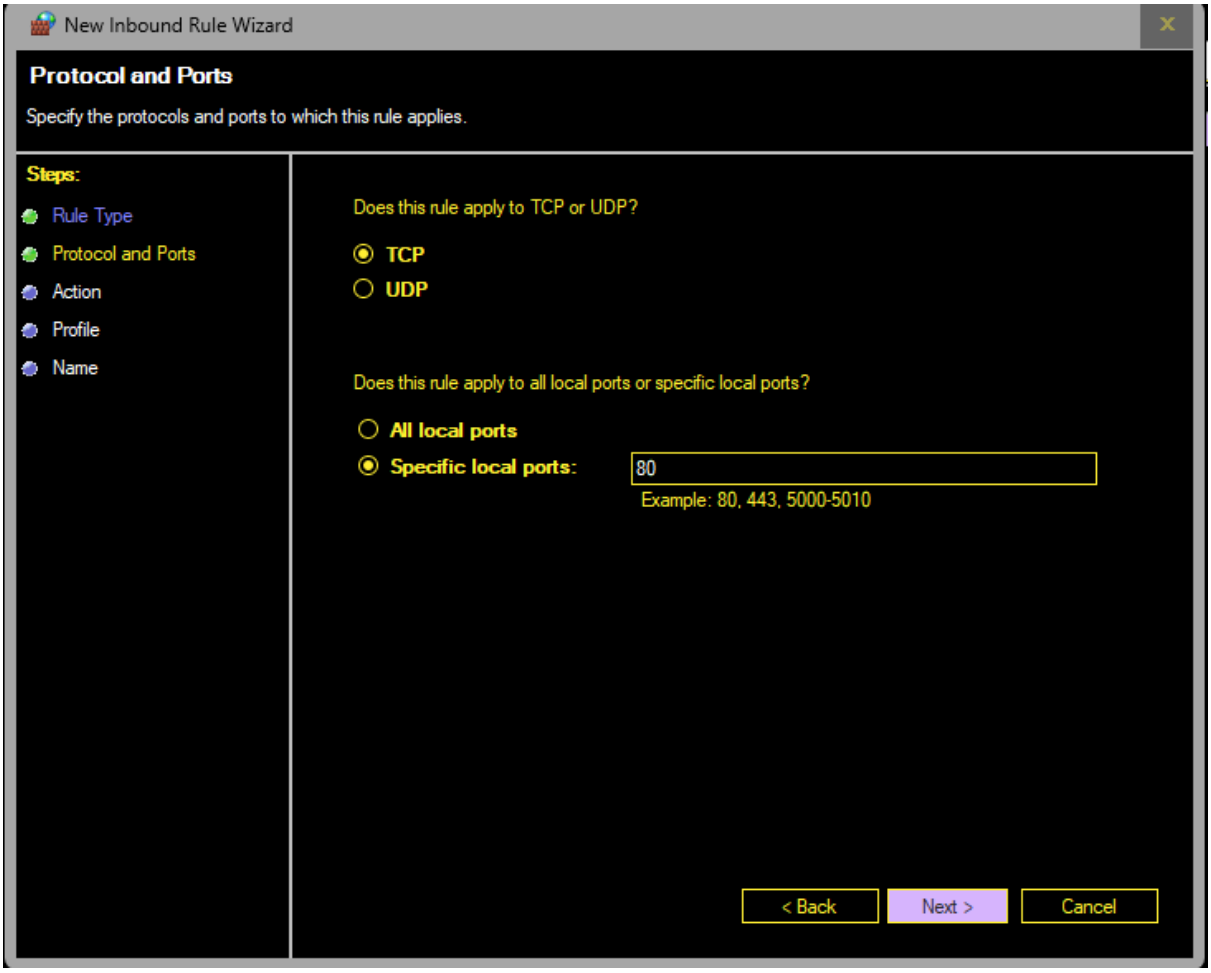
Practical No. 6

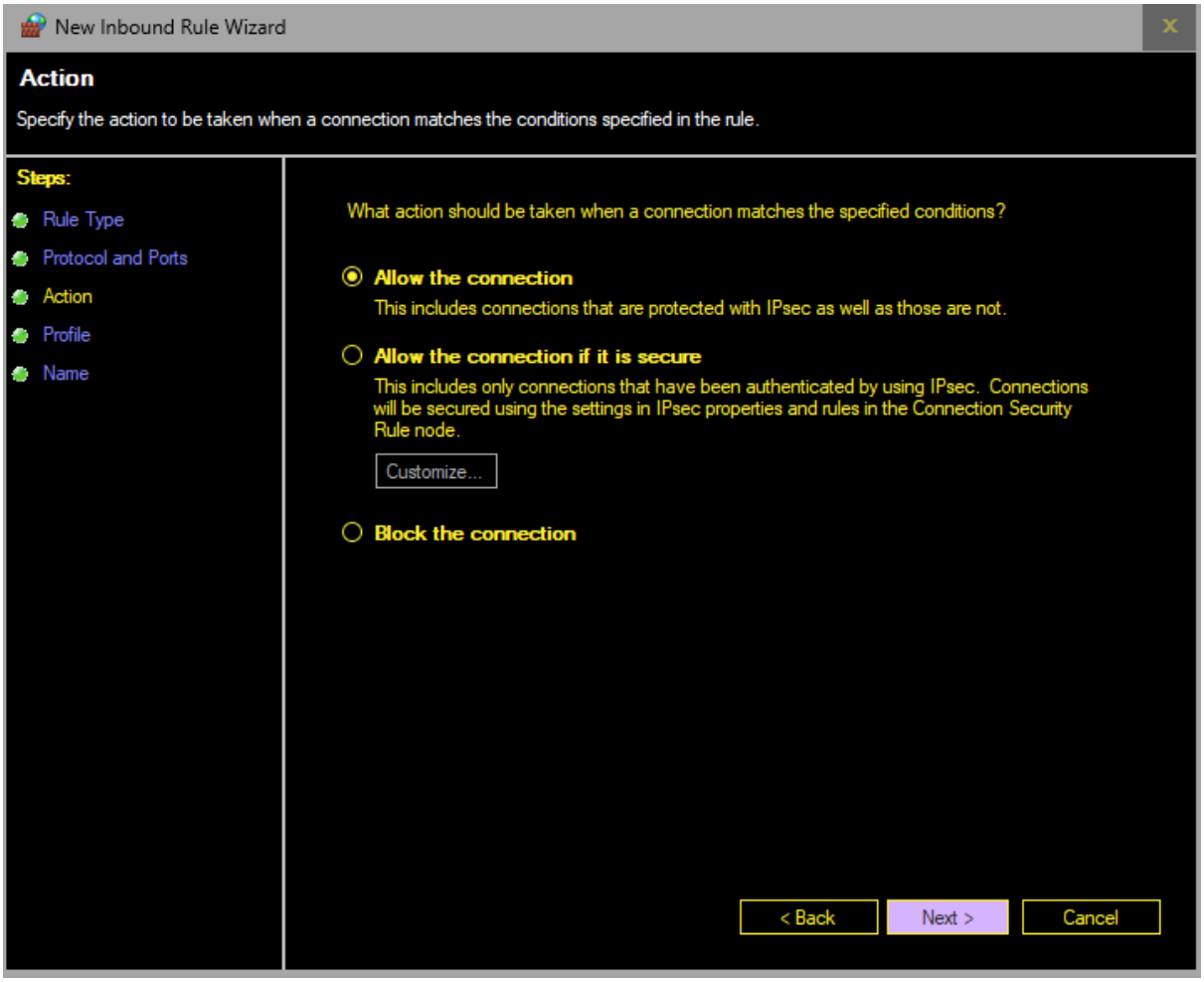
Aim: Working with docker volumes & network.

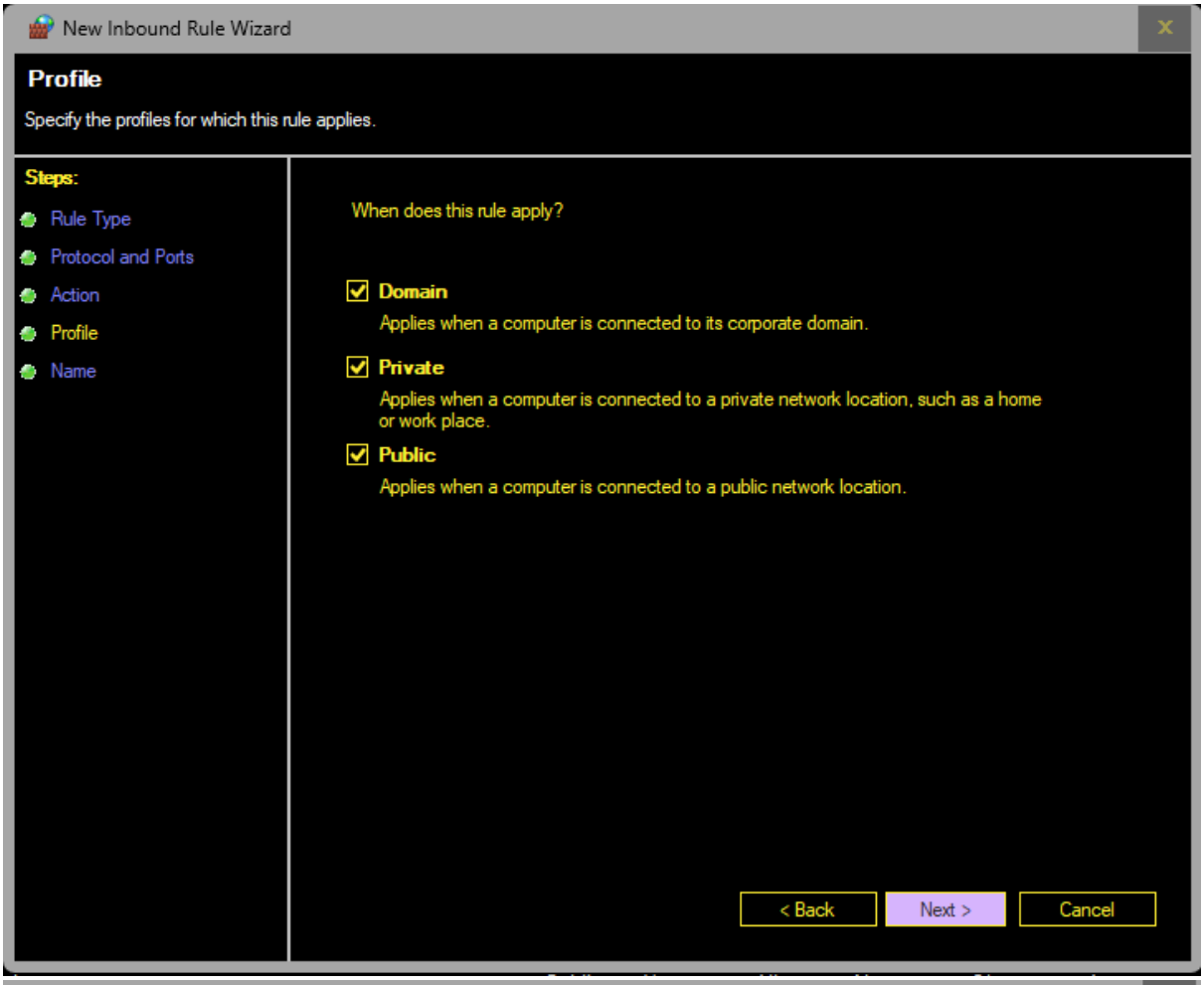
Step1: Open windows firewall with advanced security > inbound rules and create a new rule.

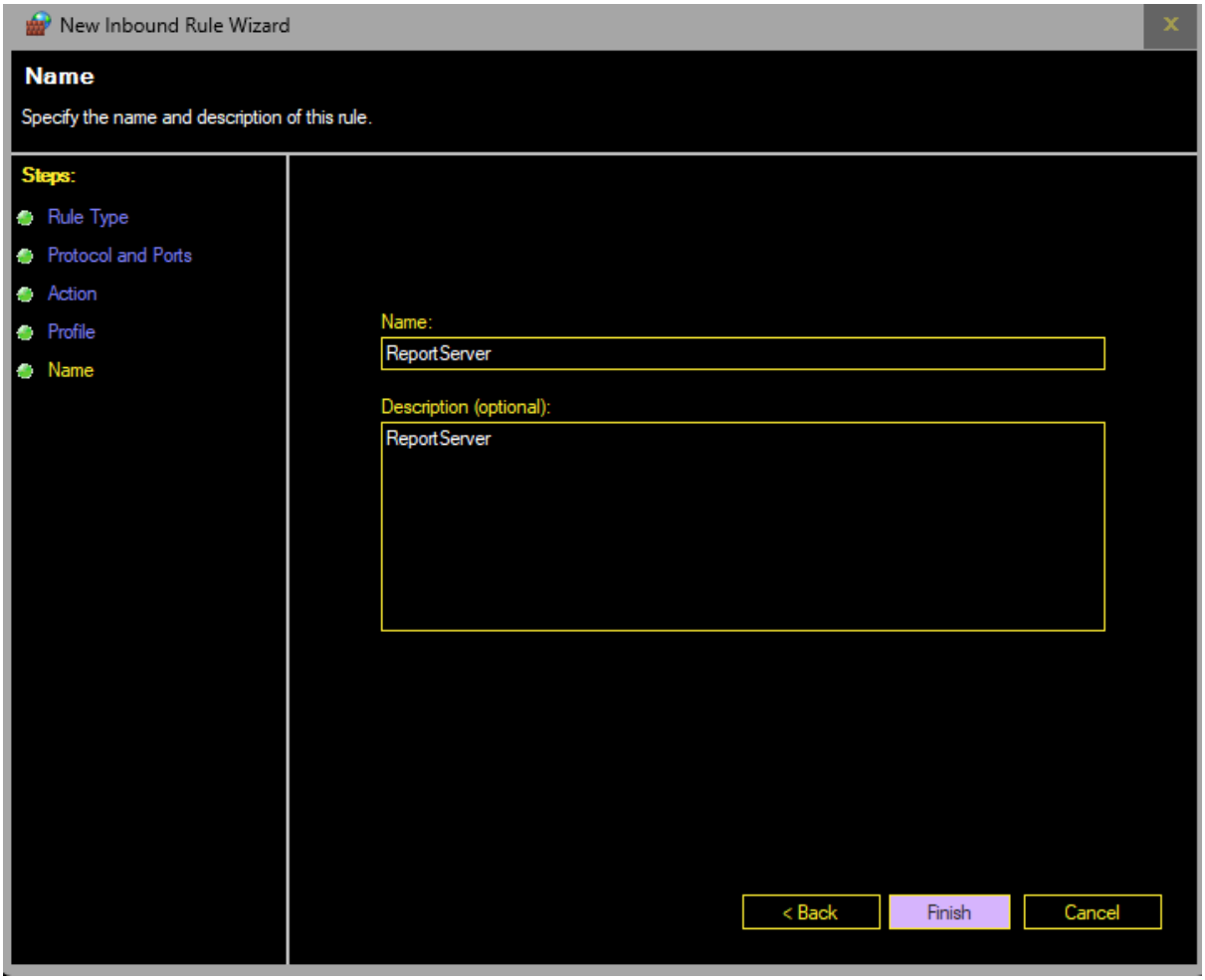


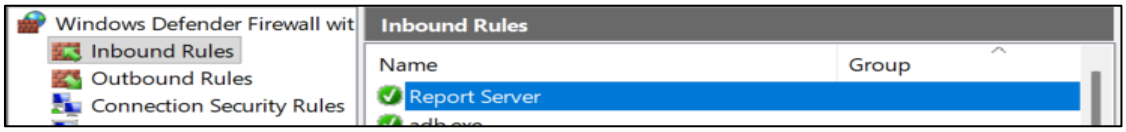



New Inbound Rule Wizard
Protocol and Ports
Specify the protocols and ports to which this rule applies.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
Does this rule apply to TCP or UDP?
TCP
UDP
Does this rule apply to all local ports or specific local ports?
All local ports
Specific local ports:
80
Example: 80, 443, 5000-5010
< Back
Next >
Cancel

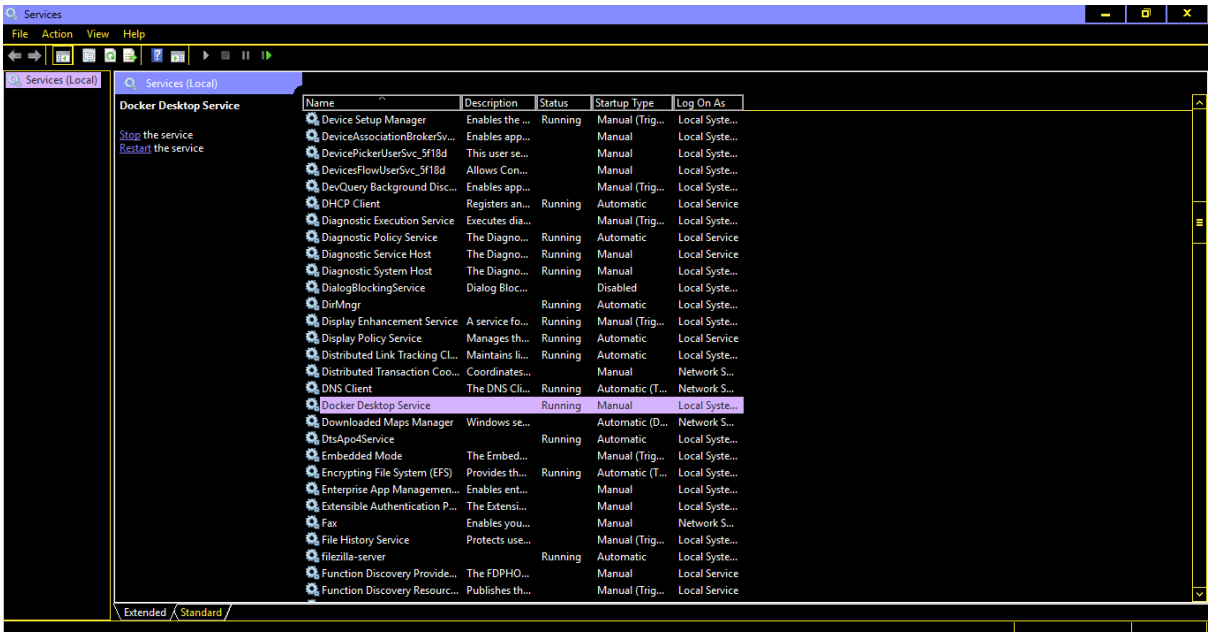

New Inbound Rule Wizard
Action
Specify the action to be taken when a connection matches the conditions specified in the rule.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
What action should be taken when a connection matches the specified conditions?
Allow the connection
This includes connections that are protected with IPsec as well as those are not.
Allow the connection if it is secure
This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.
Customize...
Block the connection
< Back
Next >
Cancel


New Inbound Rule Wizard
Profile
Specify the profiles for which this rule applies.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
When does this rule apply?
Domain
Applies when a computer is connected to its corporate domain.
Private
Applies when a computer is connected to a private network location, such as a home or work place.
Public
Applies when a computer is connected to a public network location.
< Back
Next >
Cancel

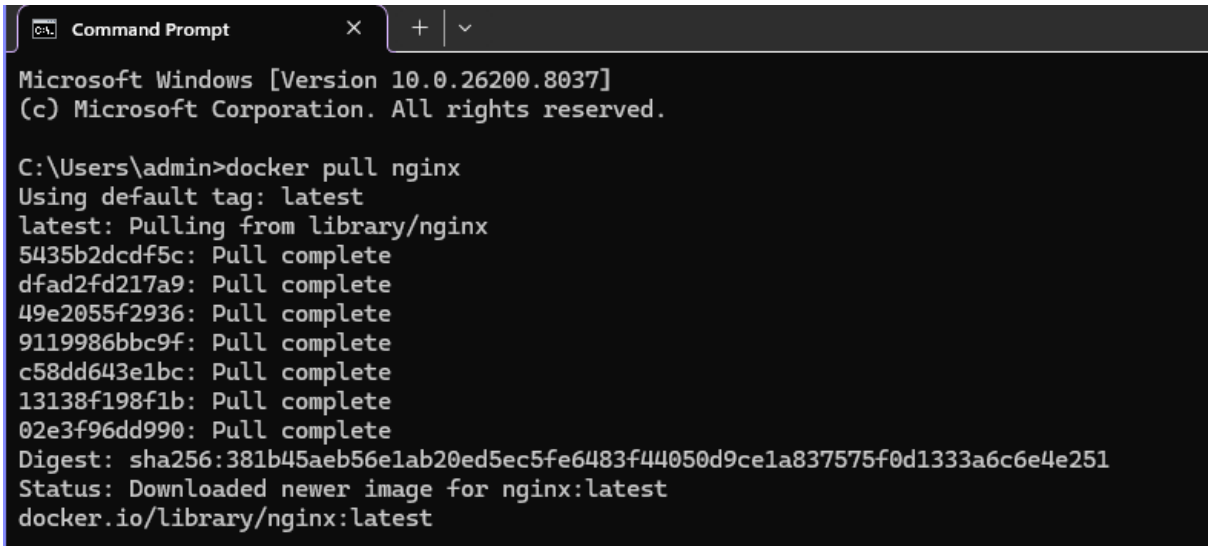

New Inbound Rule Wizard
Name
Specify the name and description of this rule.
Steps:
Rule Type
Protocol and Ports
Action
Profile
Name
Name:
ReportServer
Description (optional):
ReportServer
< Back
Finish
Cancel



Step 2: In system services, start the docker desktop services.



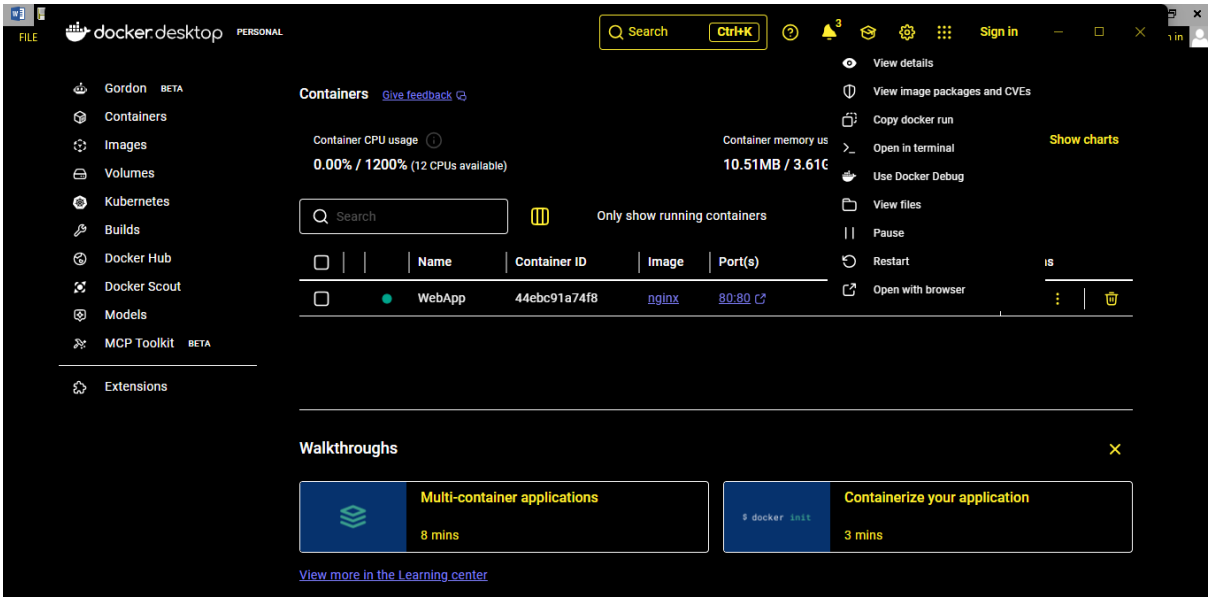
Step 3: Download the latest Nginx image from the Docker Hub registry



Step 4: Create and start a new container named "WebApp" that runs in the background, mapping your computer's port 80 to the container's port 80.

```
C:\Users\admin>docker run -it --name=WebApp -d -p 80:80 nginx
44ebc91a74f833db601a5478df271f74d53e88c95f59e3f94fad90ed32c2ebab
```

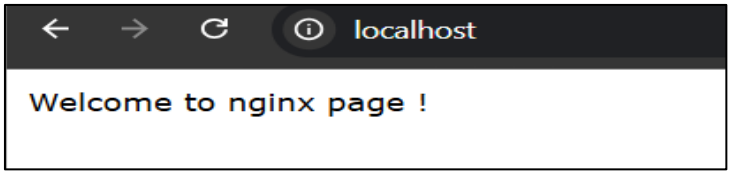
Then Go to the docker desktop and see that webapp like it has been created.



Step 5: Open an interactive bash shell inside this running WebApp container and overwrite the index.html file in the location for Nginx web files.

```
C:\Users\admin>docker exec -it WebApp bash
root@44ebc91a74f8:/# cd/usr/share/nginx/html
bash: cd/usr/share/nginx/html: No such file or directory
root@44ebc91a74f8:/# cd/usr/share/nginx/html
bash: cd/usr/share/nginx/html: No such file or directory
root@44ebc91a74f8:/# cd /usr/share/nginx/html
root@44ebc91a74f8:/usr/share/nginx/html# echo "Hello elcome to updated nginx Page" > index.html
root@44ebc91a74f8:/usr/share/nginx/html# exit
exit
```

Step 6: Open localhost:80 in browser.



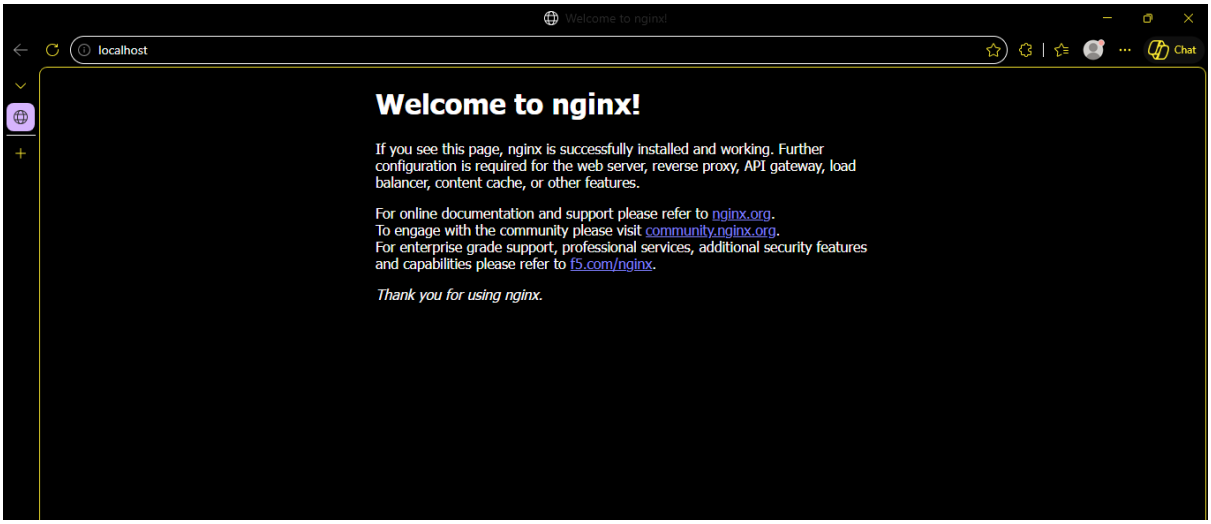
Step 7: List all currently running containers

```
C:\Users\admin>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS
 NAMES
44ebc91a74f8   nginx     "/docker-entrypoint.…"   10 minutes ago   Up 10 minutes   0.0.0.0:80->80/tcp, [::]:80->80/tcp
 WebApp
```

Step 8: Stop the previous WebApp and Create another container in Docker: WebApp1

```
C:\Users\admin>docker stop WebApp
WebApp

C:\Users\admin>docker run -it --name=WebApp1 -d -p 80:80 nginx
a8f074f91e17821184fa17eb6966e7c935cdd8d9829033f5c525d78cdb20e8e6
```

Output: the welcome page of nginx should be visible




```
C:\Users\admin>docker stop WebApp1
WebApp1
```

Volumes:

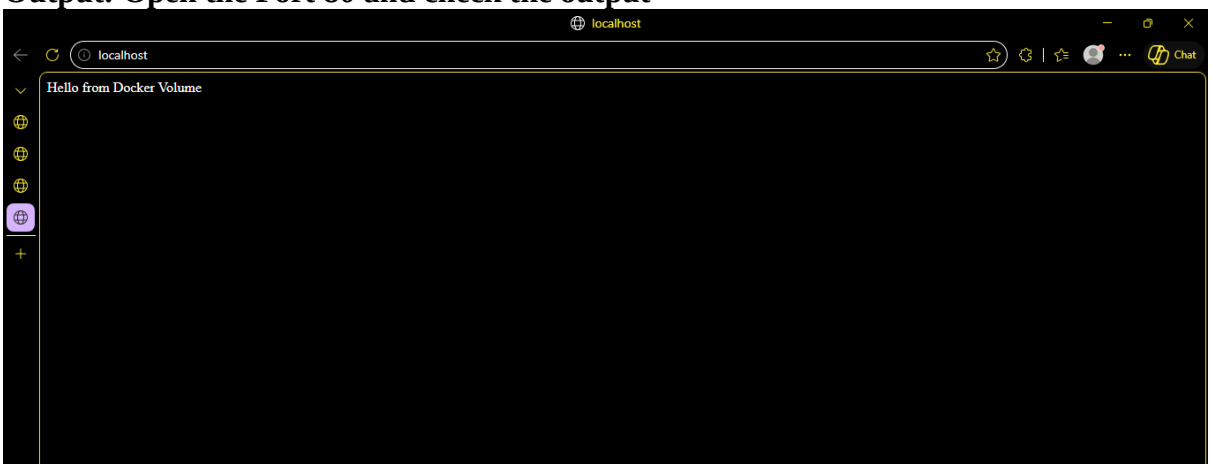
Step 9: Creation of volume (MyVolume)

```
C:\Users\admin>docker volume create MyVolume
MyVolume

C:\Users\admin>docker volume ls
DRIVER    VOLUME NAME
local     MyVolume

C:\Users\admin>docker volume inspect MyVolume
[
    {
        "CreatedAt": "2026-04-07T09:48:13Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/MyVolume/_data",
        "Name": "MyVolume",
        "Options": null,
        "Scope": "local"
    }
]

C:\Users\admin>docker stop WebApp1
WebApp1
```

Step 10: Create a Container (WebApp3) inside the container MyVolume and Open an interactive bash shell inside this running WebApp container and overwrite the index.html file

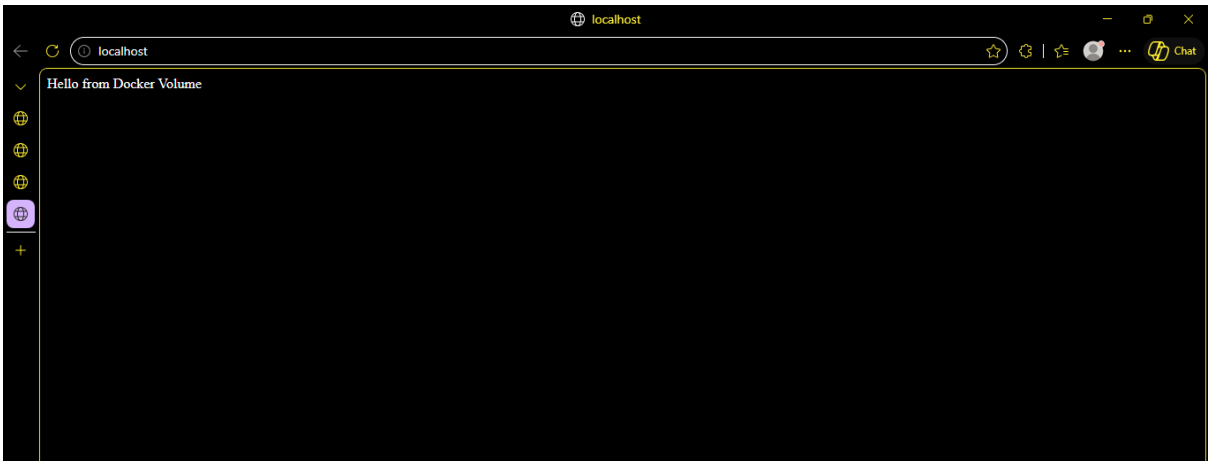
```
C:\Users\admin>docker run -d --name=WebApp2 --mount source=MyVolume,destination=/usr/share/nginx/html -p 80:80 nginx
32b2011858f014cb453a61f5a8911c4a808db59ecfb696e75cbe0a1404bbc9c8

C:\Users\admin>docker exec -it WebApp2 bash
root@32b2011858f0:/# cd /usr/share/nginx/html
root@32b2011858f0:/usr/share/nginx/html# ls
50x.html  index.html
root@32b2011858f0:/usr/share/nginx/html# echo "Hello from Docker Volume" > index.html
root@32b2011858f0:/usr/share/nginx/html# exit
exit
```

Output: Open the Port 80 and check the output



Step 11: Stop the above container (WebApp3) and create another container (WebApp4) within the volume (MyVolume)

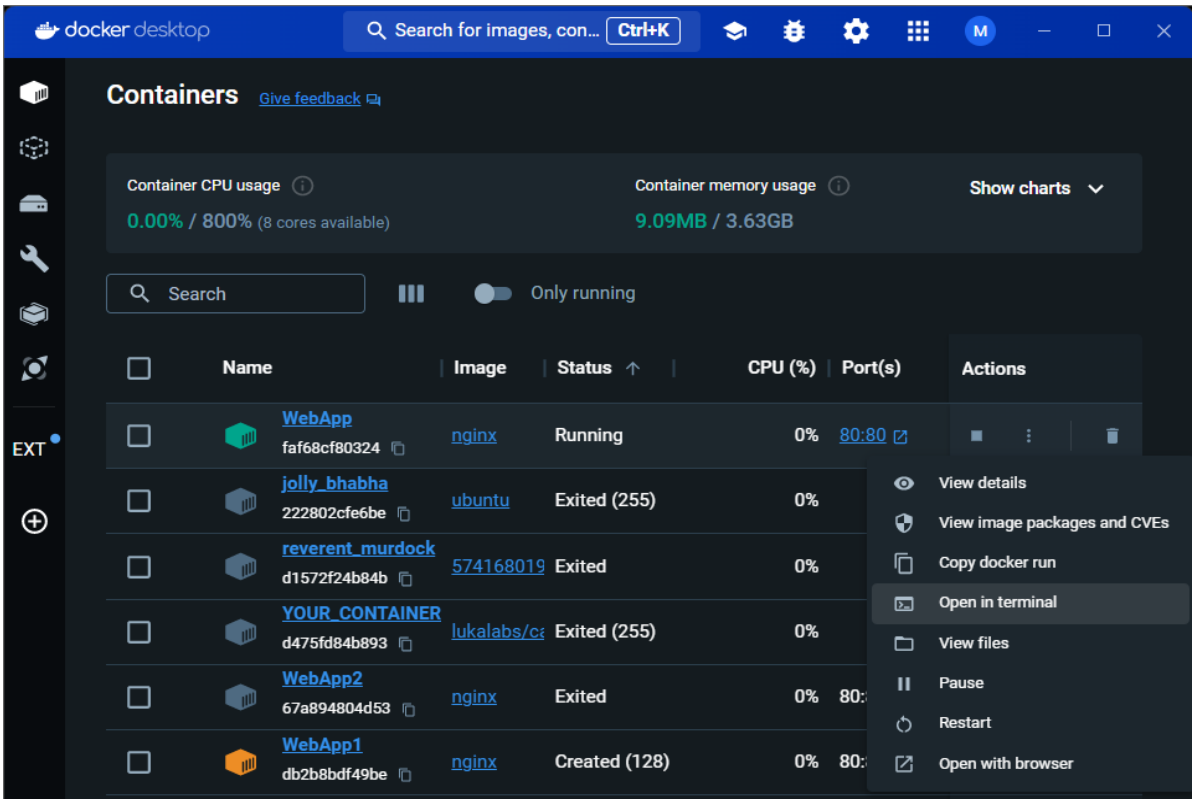
```
C:\Users\admin>docker stop WebApp2
WebApp2

C:\Users\admin>docker run -d --name=WebApp4 --mount source=MyVolume,destination=/usr/share/nginx/html -p 80:80 nginx
498b0058448ef0cc75de817dbc4fc2f25701adb46ff4863c07e8e18e3300066c

C:\Users\admin>
```

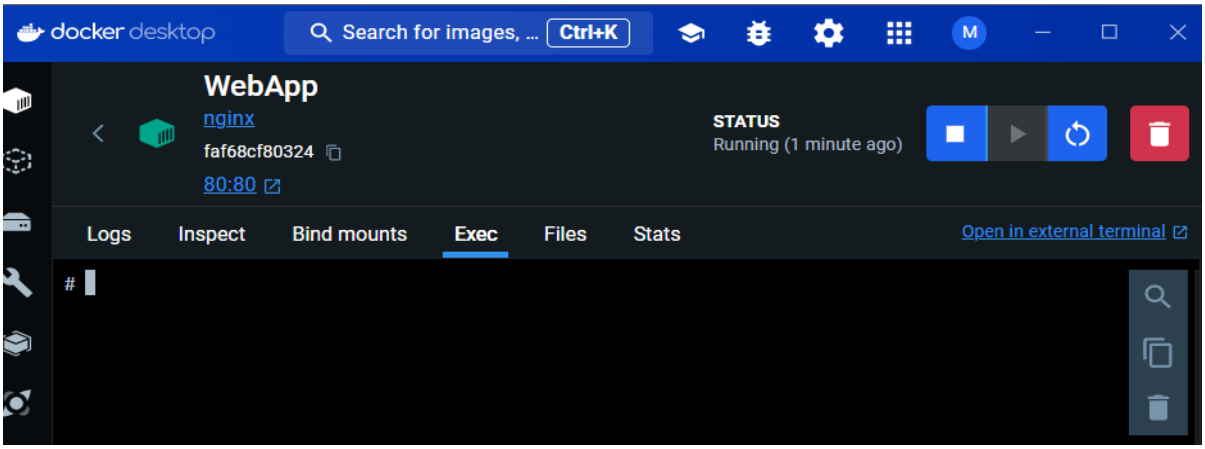


Practical No. 7

Aim: Installing software packages in docker container.

Step 1 – Open the container in terminal from Docker Desktop





Step 2: Update the OS.

Logs Inspect Bind mounts Exec Files Stats Open in external terminal

```
# apt get -y update
Error: Invalid operation get
# apt-get update
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [119 kB]
Fetched 10.0 MB in 3s (3131 kB/s)
Reading package lists... Done
#
```

Step 3: Install firefox and vim.

```
# apt-get install vim
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libgpm2 libsodium23 vim-common vim-runtime xxd
Suggested packages:
  gpm ctags vim-doc vim-scripts
The following NEW packages will be installed:
  libgpm2 libsodium23 vim vim-common vim-runtime xxd
0 upgraded, 6 newly installed, 0 to remove and 0 not upgraded.
Need to get 9496 kB of archives.
After this operation, 45.9 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://deb.debian.org/debian trixie/main amd64 vim-common all 2:9.1.1230-2 [421 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libgpm2 amd64 1.20.7-11+b2 [14.4 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 libsodium23 amd64 1.0.18-1+deb13u1 [165 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 vim-runtime all 2:9.1.1230-2 [7127 kB]
```

```
update-alternatives: warning: skip creation of /usr/share/man/tr/man1/view.1.gz because associated file
 /usr/share/man/tr/man1/vim.1.gz (of link group view) doesn't exist
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/vim (vim) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/vimdiff (vimdiff) in auto mode
Processing triggers for libc-bin (2.41-12+deb13u2) ...
#
```

```
# apt-get install firefox-esr
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  adduser adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core dbus
  dbus-bin dbus-daemon dbus-session-bus-common dbus-system-bus-common dbus-user-session
  dconf-gsettings-backend dconf-service dmsetup fontconfig gsettings-desktop-schemas
  gtk-update-icon-cache hicolor-icon-theme i965-va-driver intel-media-va-driver libapparmor1
  libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk1.0-0t64 libatomic1 libatspi2.0-0t64
  libavahi-client3 libavahi-common-data libavahi-common3 libavcodec61 libavutil59 libcairo-gobject2
```

```
update-alternatives: warning: skip creation of /usr/share/man/man1/gnome-www-browser.1.gz because assoc
iated file /usr/share/man/man1/firefox-esr.1.gz (of link group gnome-www-browser) doesn't exist
Setting up libgtk-3-bin (3.24.49-3) ...
Setting up dmsetup (2:1.02.205-2) ...
Setting up libdevmapper1.02.1:amd64 (2:1.02.205-2) ...
Setting up libcryptsetup12:amd64 (2:2.7.5-2) ...
Setting up systemd-cryptsetup (257.9-1~deb13u1) ...
Processing triggers for libc-bin (2.41-12+deb13u2) ...
Processing triggers for systemd (257.9-1~deb13u1) ...
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.12+dfsg-4) ...
#
```

Step 4: Run vim and Firefox to verify if the software package has been installed

```
# vim .
```

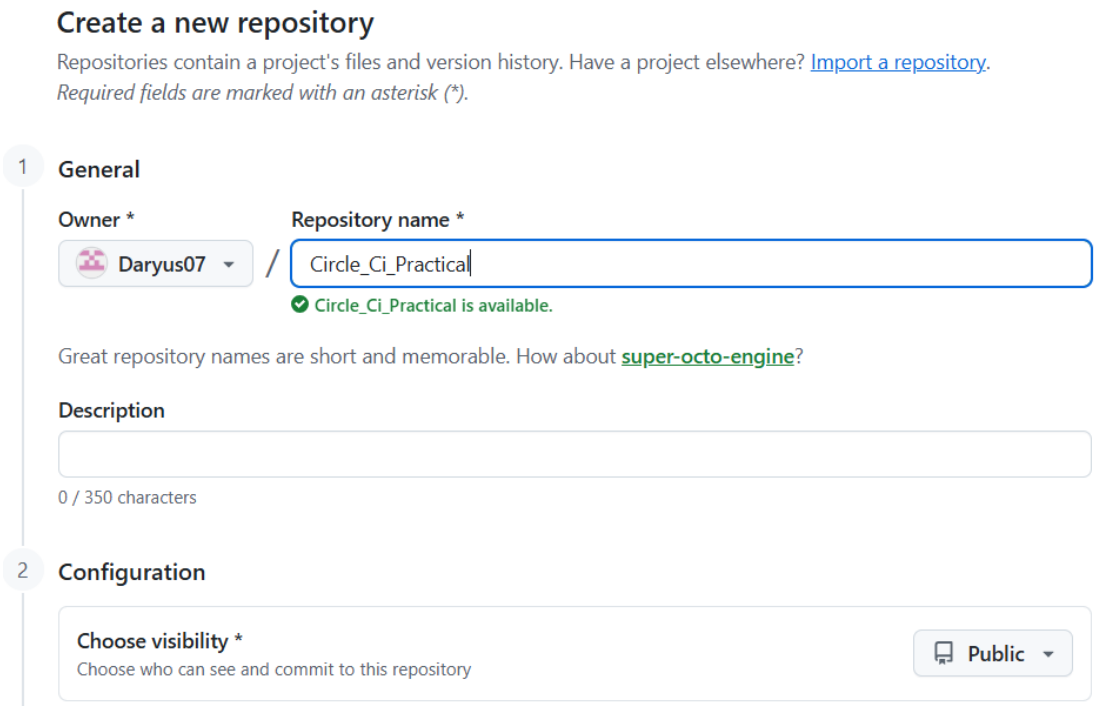
```
" ==============================================================================
" Netrw Directory Listing                                       (netrw v179)
"   /
"   Sorted by      name
"   Sort sequence: [\/]$,*,\(\.bak\|\~\|\.swp\|\.o\|\.info\|\.aux\|\.log\|\.dvi\|\.bbl\|\.blg\|\.brf\|\
"   Quick Help: <F1>:help  -:go up dir  D:delete  R:rename  s:sort-by  x:special
" ==============================================================================
./
./
boot/
dev/
docker-entrypoint.d/
etc/
home/
media/
mnt/
opt/
proc/
root/
run/
srv/
sys/
tmp/
usr/
var/
.dockerenv*
bin@      --> usr/bin
docker-entrypoint.sh*
lib64@    --> usr/lib64
lib@      --> usr/lib
sbin@     --> usr/sbin
~
~
~
~
"." is a directory                                                    8,1           All
```

```
# firefox --headless
*** You are running in headless mode.
[GFX1-]: glxtest: libpci missing
[GFX1-]: glxtest: Unable to open a connection to the X server
[GFX1-]: No GPUs detected via PCI

[GFX1-]: RenderCompositorSWGL failed mapping default framebuffer, no dt
```

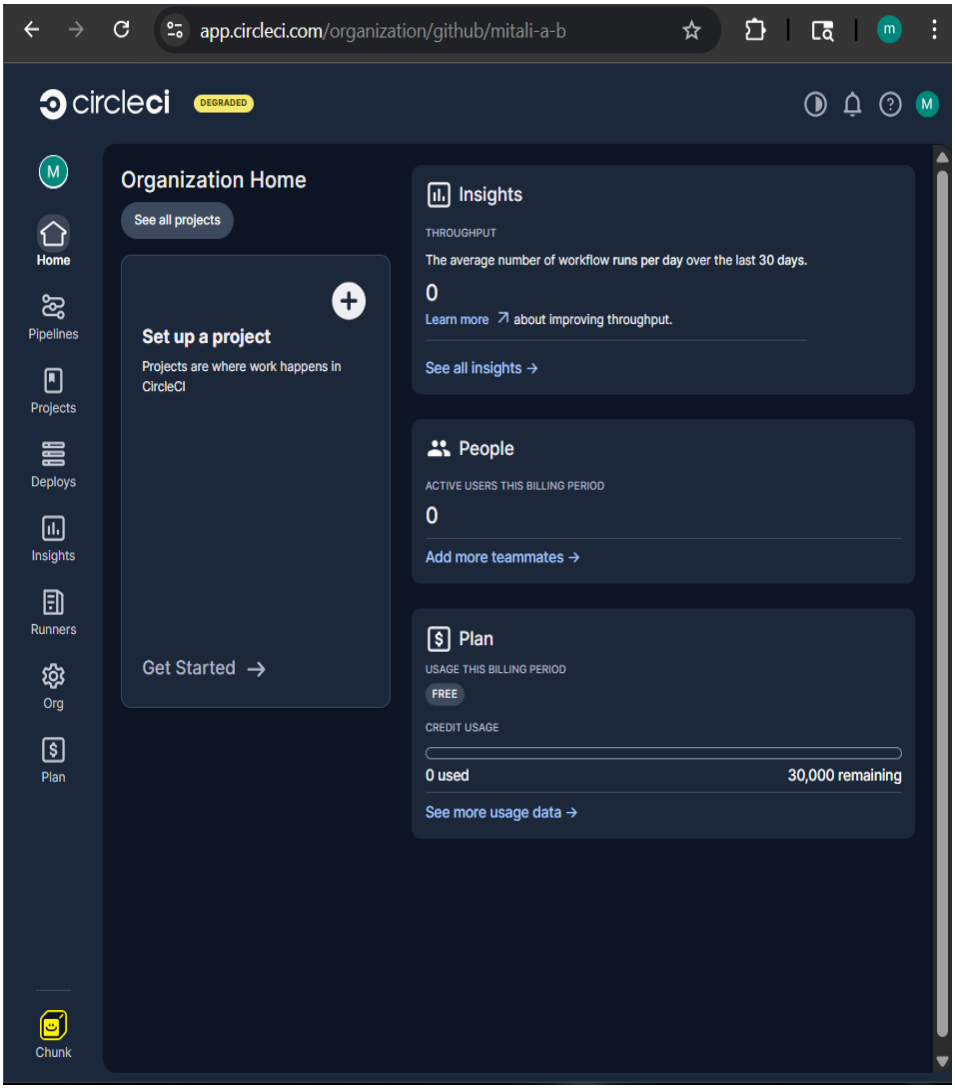
Practical No. 8

Aim: To implement Continuous Integration using CircleCI by connecting a GitHub repository and automating build and test processes.

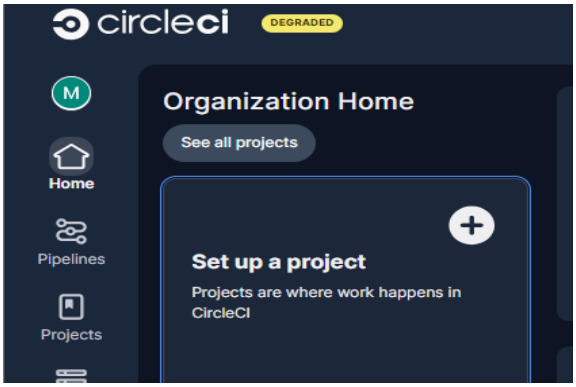
Step 1: Create a Repository on GitHub.

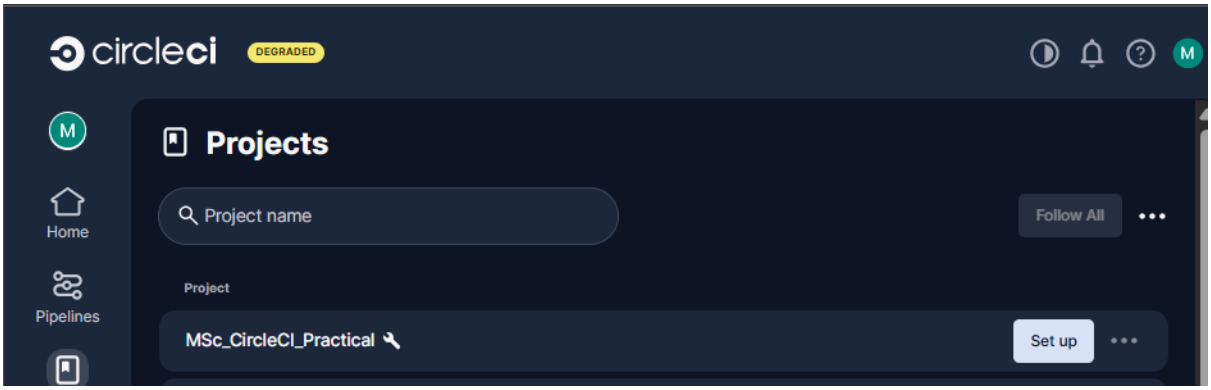


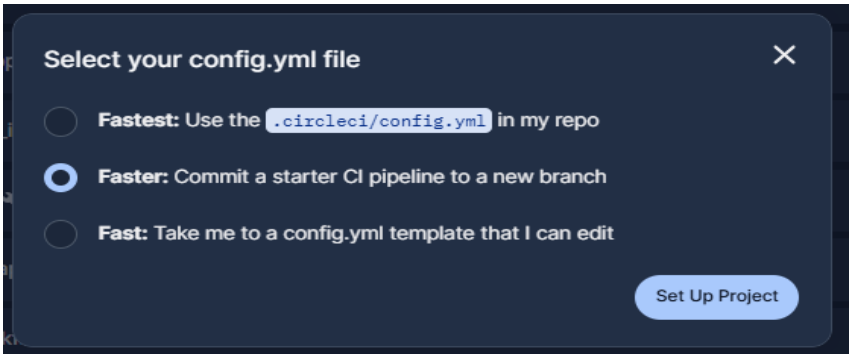
Step 2: Sign in to CircleCI - https://app.circleci.com/ with GitHub.

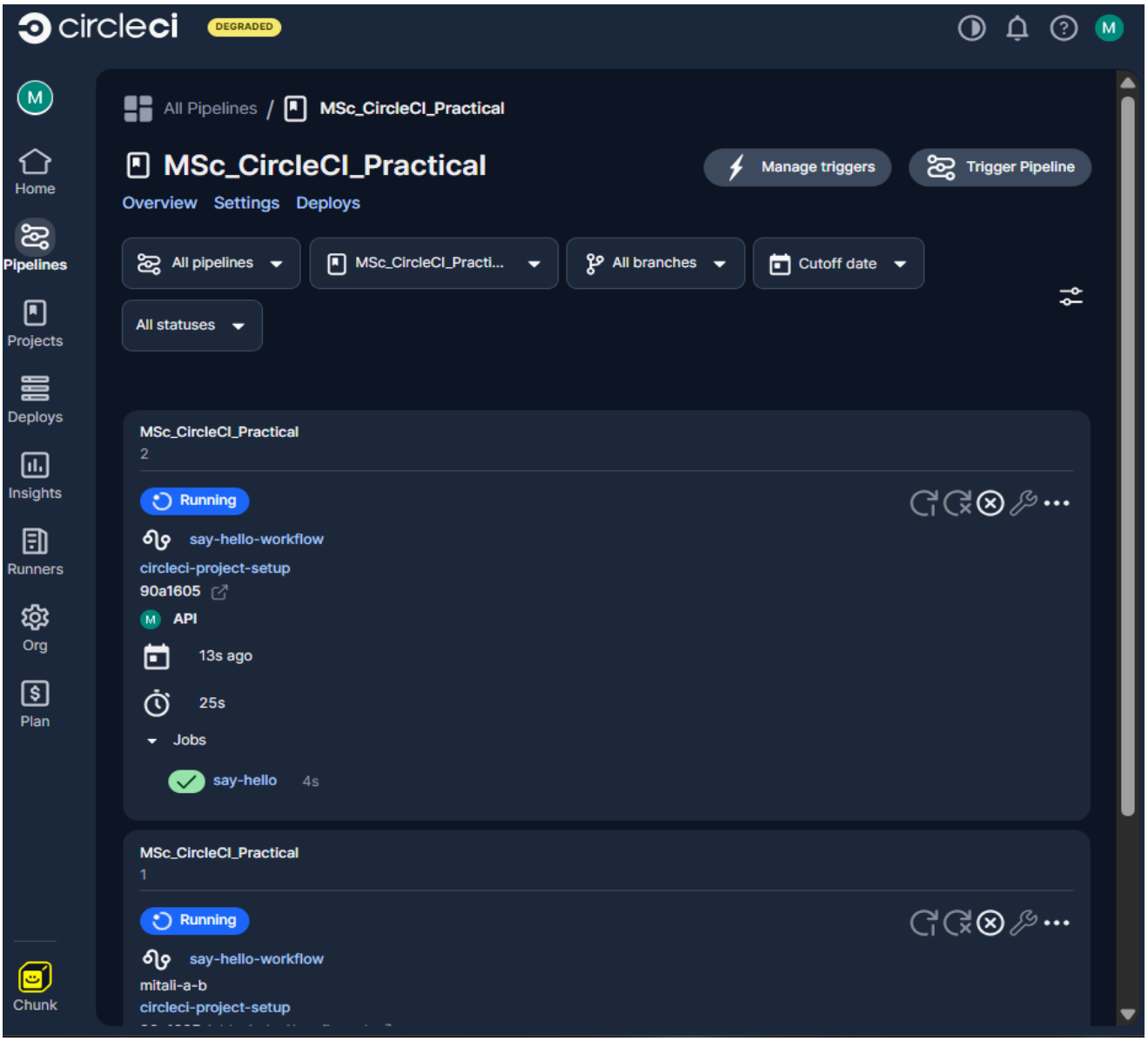


Step 3: Add Project in CircleCI

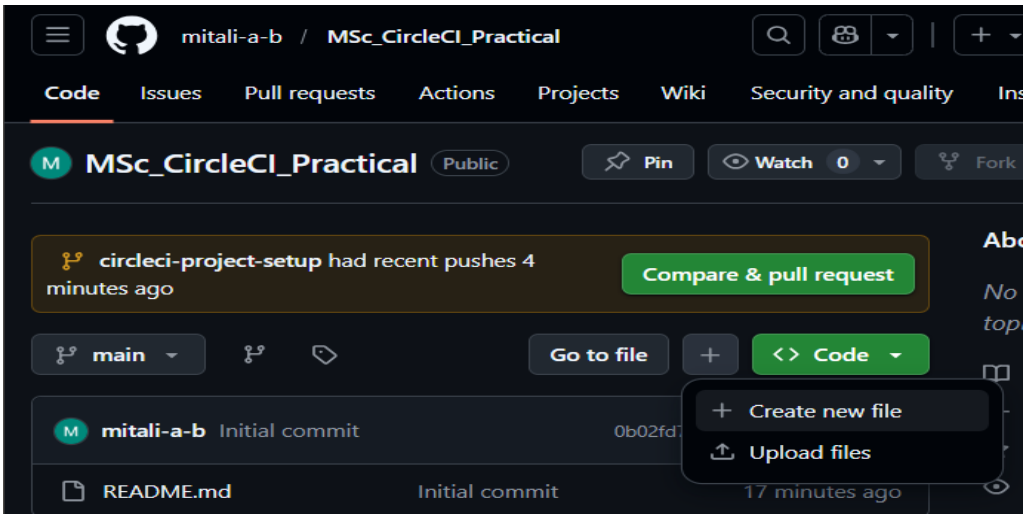


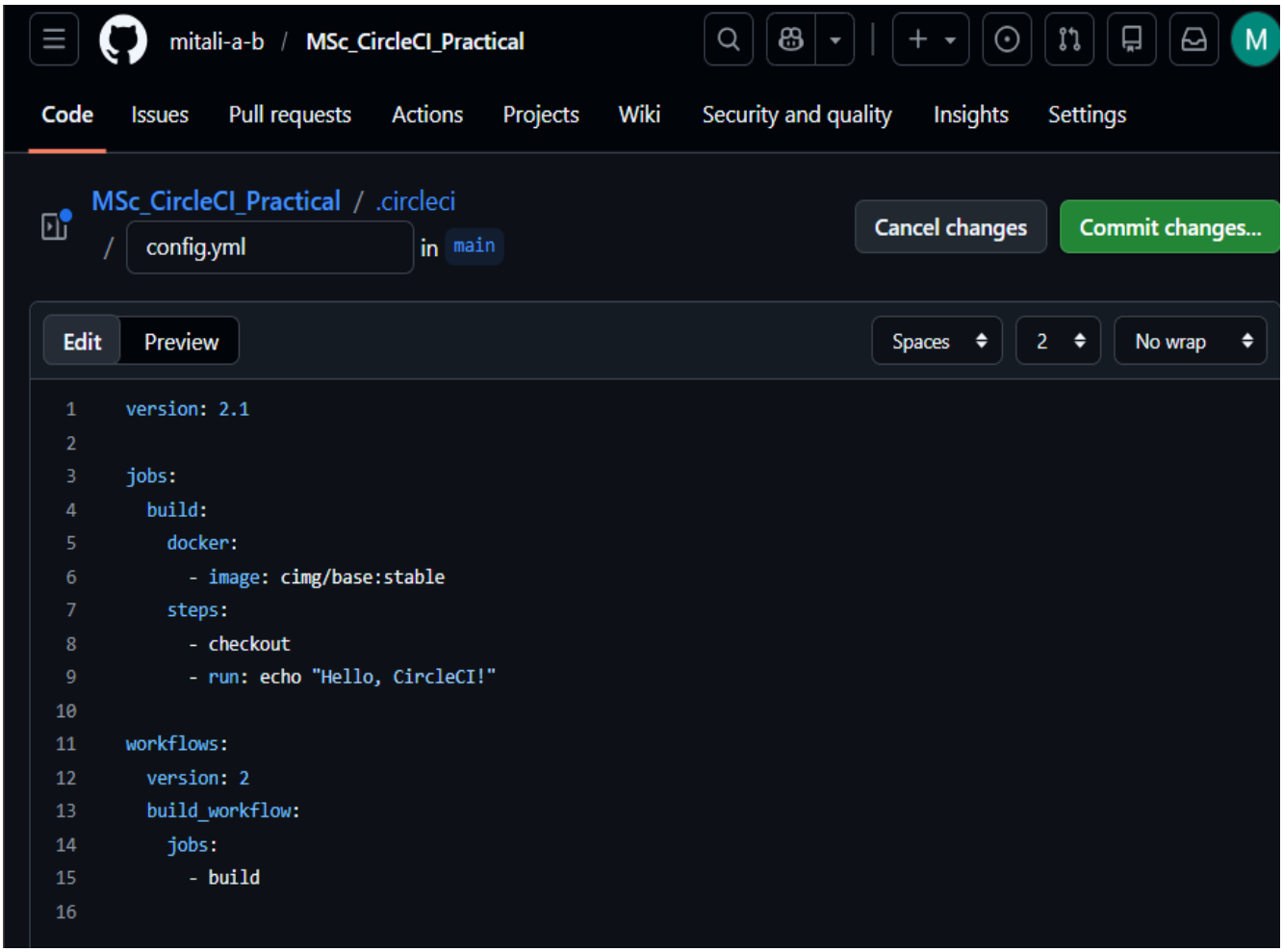

circleci
DEGRADED
Projects
Home
Pipelines
Project name
Follow All
Project
MSc_CircleCI_Practical
Set up


Select your config.yml file
Fastest: Use the .circleci/config.yml in my repo
Faster: Commit a starter CI pipeline to a new branch
Fast: Take me to a config.yml template that I can edit
Set Up Project

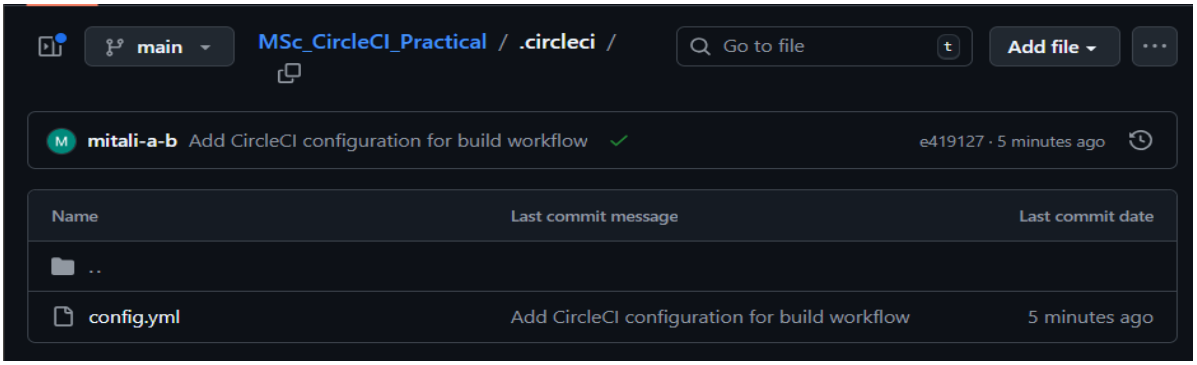
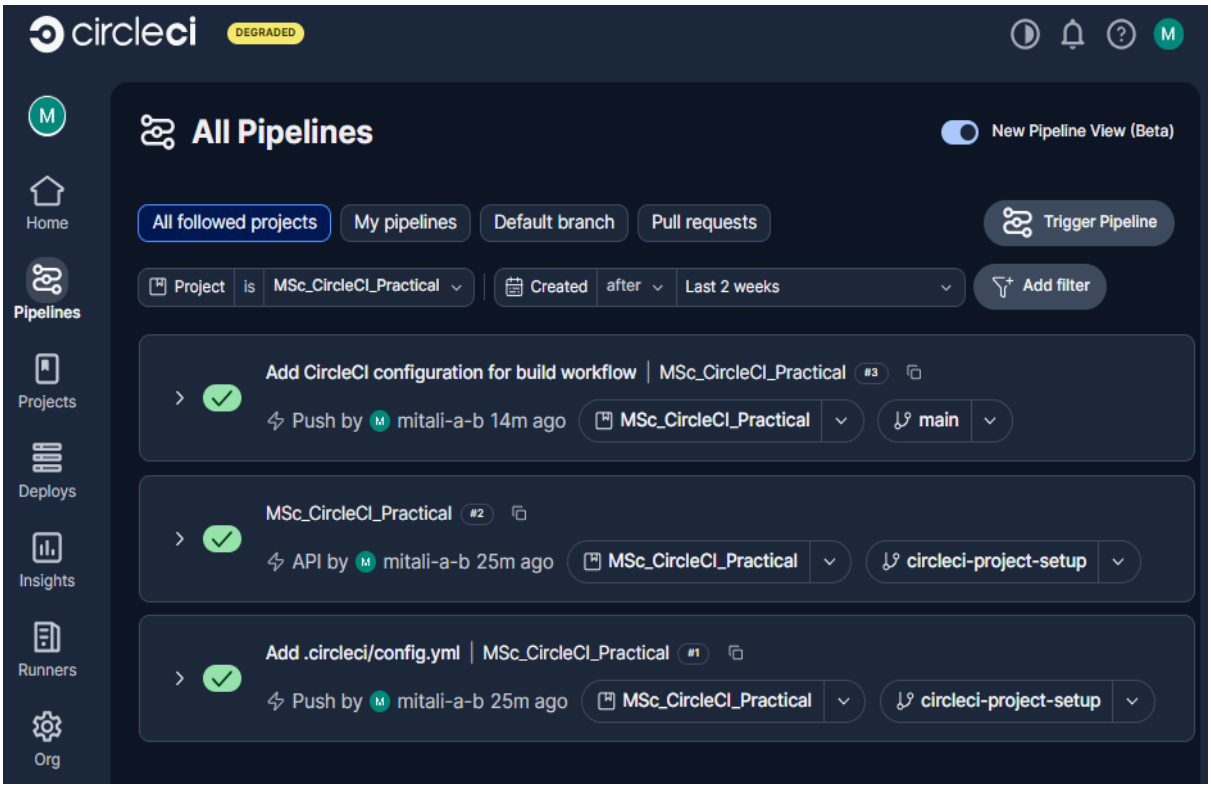
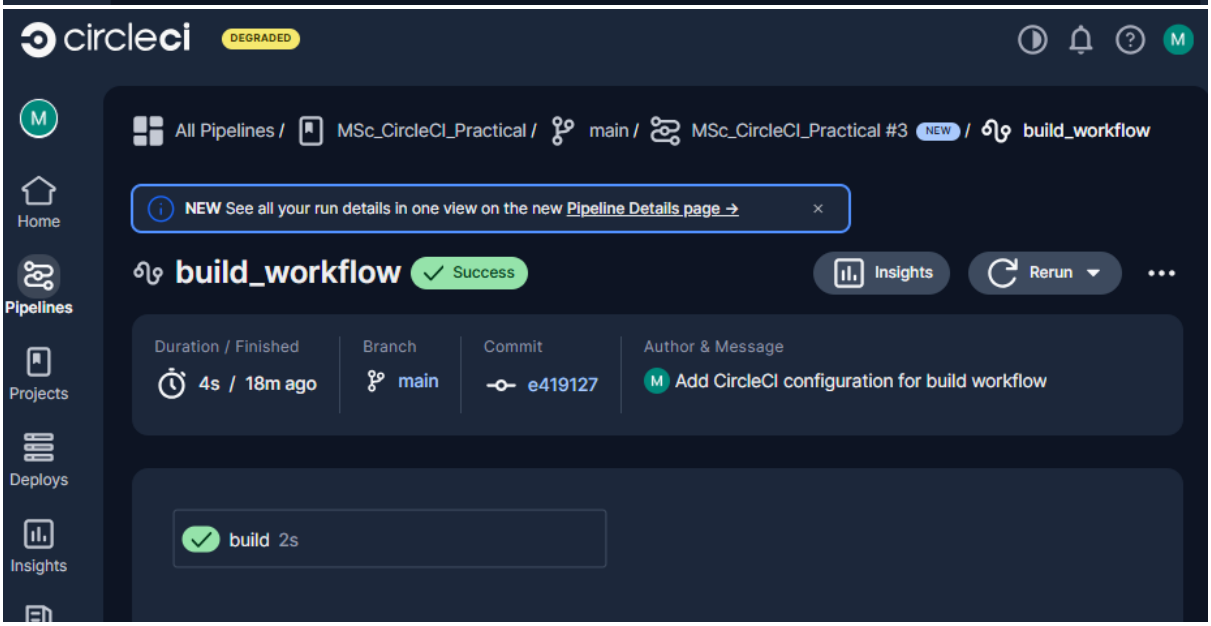

circleci
DEGRADED
All Pipelines / MSc_CircleCI_Practical
MSc_CircleCI_Practical
Manage triggers
Trigger Pipeline
Overview Settings Deploys
All pipelines
MSc_CircleCI_Practi...
All branches
Cutoff date
All statuses
Home
Pipelines
Projects
Deploys
Insights
Runners
Org
Plan
Chunk
MSc_CircleCI_Practical
2
Running
say-hello-workflow
circleci-project-setup
90a1605
API
13s ago
25s
Jobs
say-hello 4s
MSc_CircleCI_Practical
1
Running
say-hello-workflow
mitali-a-b
circleci-project-setup

Step 4: Create Configuration File in the repo in github. Create folder .circleci and inside it create a config.yml and Add Basic Configuration in it.





Commit changes



Step 5: Run Pipeline - Go to CircleCI dashboard → Pipeline will start automatically.